



from the labs

Going faster

Theoretically, a scramjet engine can reach speeds up to Mach 24, eight times faster than a conventional aircraft

The Little Engines That May

The engines have always been there, but now, scientists are looking at ways to make them save us in the 21st century



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For so many years, it was easy to make advances in transportation. To go faster, you need to either (a) burn more fuel, (b) burn it hotter, (c) burn it faster, or (d) all of the above. That's how one-cylinder engines became V12s, how V12s became jet engines, and how jet engines became rockets.

But the old ways won't work any more. 'Tis a new world, that demands new things — we must move faster, but we must also do so without burning up fossil fuels. Some of us will settle for just moving, as long as we know we aren't polluting the atmosphere. Then there's air travel — we need to spend less time in the air, but still travel great distances. We must ride on the edge of space, but without the big, expensive multi-stage rockets. For every one of our desires, there's something holding us back.

It may seem like that scientists would have a lot of back-to-the-drawing-board-ing to do, but the good news is that the engines that will take us into the future have already existed for ages.

...And they breathe air

Thus far, the most popular way to get to the edge of the atmosphere is to use a rocket. But, of course, it isn't that simple. For even the smallest thing, we need massive rocket engines, which (apart from taking up a lot of space) aren't the most desirable way to get from point A to point B. It isn't enough to stuff these rockets with fuel — we also need to stuff them with an oxidiser that supplies the oxygen to burn the fuel.

So if we are to fly high, we must first expend with these massive rockets, and bring back engines that go back as far as 1913. They're called air-breathing engines, because instead of using a separate oxidiser, they use oxygen from the air to burn the fuel.

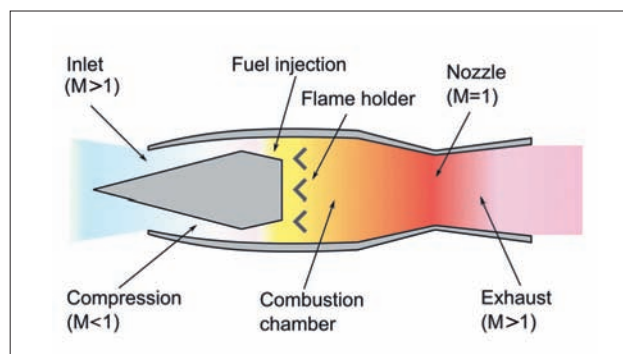
Like all engines, the basic principle of the air-breathing engine is the same — compress air, throw in some fuel, let it burn, enjoy, repeat. It's how the air gets there that makes the difference.

The ramjet

This engine works on a simple principle — air enters a chamber at high speed, is compressed, and then is squished out of a tiny hole at the other end, which causes it to be expelled with even more pressure, propelling the engine forward.

Now, if you stuck some fuel at the point where the air was most compressed and set it on fire, you get an explosion of hot air. When that explosion goes out the other end, you get propulsion that demands the name "jet". Which is probably why the creator called it a ramjet. Not only is it a powerhouse, the ramjet is pretty efficient at supersonic speeds.

For all its goodness, though, the ramjet still won't be the engine that takes us into the future. We need aircraft that can



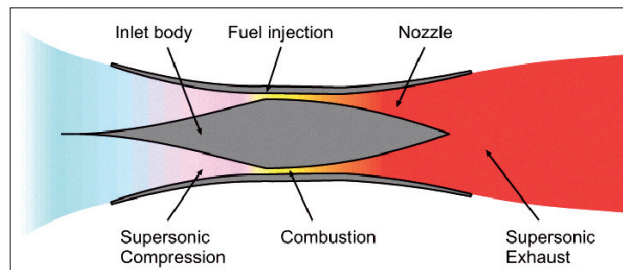
How ramjets work. The pointy bit at the front creates its own shockwave of air, and sends it into the combustion chamber behind it, where it's mixed with fuel, set on fire, and sent out the other end.

travel at ridiculous speeds — Mach 10 or so — but the ramjet stops being useful at around Mach 5. When the air compresses inside the jet, its temperature rises, and above Mach 5, the air gets too hot to burn. So someone tweaked the design.

The scramjet

Where the ramjet stops, the scramjet begins. It's short for Supersonic Combustion Ramjet, and like the name suggests, it's designed to operate at several times the speed of sound. The improvement on the ramjet is simple enough: don't compress the air so much that it overheats, and toss the fuel into the mix a bit earlier. The result is a sleeker, shorter version of the ramjet.

At relatively lower speeds (Mach 3 or so), the scramjet will work like any regular ramjet, making it the engine of choice for future aircraft. In 2004, NASA's X-43A took its first successful stab at hypersonic flight, and reached Mach 9.6. In February, a UK firm called Reaction Engines received a million Euros (about six crores) to develop the Skylon, an unmanned scramjet that



The scramjet does away with the relatively large combustion chamber: the fuel is mixed with the air almost as soon as it gets in, and the hot air blows out the back.